GDP per Capita and Corruption: Do Additional Macroeconomic Controls Matter?

AI Economist*

Abstract

A one standard deviation rise in log GDP per capita predicts a 0.70 standard deviation increase in the Corruption Perceptions Index, and no other macroeconomic indicator meaningfully improves this prediction. We estimate this relationship using a panel of 150 countries over 2012–2024 with year fixed effects, then test whether 11 additional controls improve out-of-sample accuracy via regularized regression (LASSO, Ridge, Elastic Net). The GDP-only model explains 49.9% of CPI variance. Adding controls does not reduce cross-validated RMSE (21.67 vs. 21.76). LASSO selects only 6 of 12 candidates; GDP carries the dominant coefficient (19.42). Nominal GDP per capita marginally outperforms PPP-adjusted GDP (AIC difference: −9.7), consistent with PPP adding measurement noise. Cross-country corruption perception is overwhelmingly a function of income, with limited scope for additional macroeconomic determinants.

Keywords corruption; GDP per capita; regularized regression; LASSO; PPP measurement

1 Introduction

Richer countries are less corrupt. This empirical regularity has been documented across dozens of cross-country studies spanning three decades (Mauro, 1995; Treisman, 2000; Serra, 2006), and it remains one of the most robust findings in the political economy of development. But the strength of this relationship raises a natural next question: does anything else matter? Once we know a country's income per capita, do other macroeconomic indicators—trade openness, inflation, government spending, education, natural resources, democracy—add independent information about its perceived corruption level? Or is the bivariate GDP-corruption gradient so dominant that additional controls amount to noise?

We find that the answer is overwhelmingly the latter. Using a panel of 150 countries observed annually from 2012 to 2024, we show that log GDP per capita alone explains 49.9 percent of cross-country variation in the Corruption Perceptions Index (CPI). Adding 11 macroeconomic and institutional control variables—selected from across the prior literature as candidate determinants—produces no meaningful improvement in out-of-sample predictive accuracy across four estimators (OLS, LASSO, Ridge, and Elastic Net). Regularized regression confirms this result: LASSO shrinks 6 of 12 predictors to zero, and GDP carries a coefficient (19.42) that dwarfs all other selected variables. The hypothesis that "GDP per capita is all you need" survives a battery of robustness checks, including an alternative dependent variable, income-group subsamples, and exclusion of year fixed effects. We also document a novel empirical finding: nominal (current US$) GDP per capita marginally but consistently outperforms PPP-adjusted GDP per capita in predicting corruption, supporting the claim that PPP adjustment adds measurement noise (Deaton, 2010) without improving the corruption signal.

This paper makes two contributions. First, we bring regularization methods—LASSO, Ridge, and Elastic Net—to a question that has been examined almost exclusively through OLS with hand-picked controls or Extreme Bounds Analysis (Serra, 2006; Seldadyo and de Haan, 2006). Regularization directly addresses the multicollinearity concern that plagues cross-country corruption regressions, where macroeconomic indicators tend to move together, and it provides a principled framework for out-of-sample evaluation. Second, we provide the first systematic comparison of

*AI Economist Research Pipeline. Correspondence: ai-economist@example.com. This paper was generated by an automated economics research system.

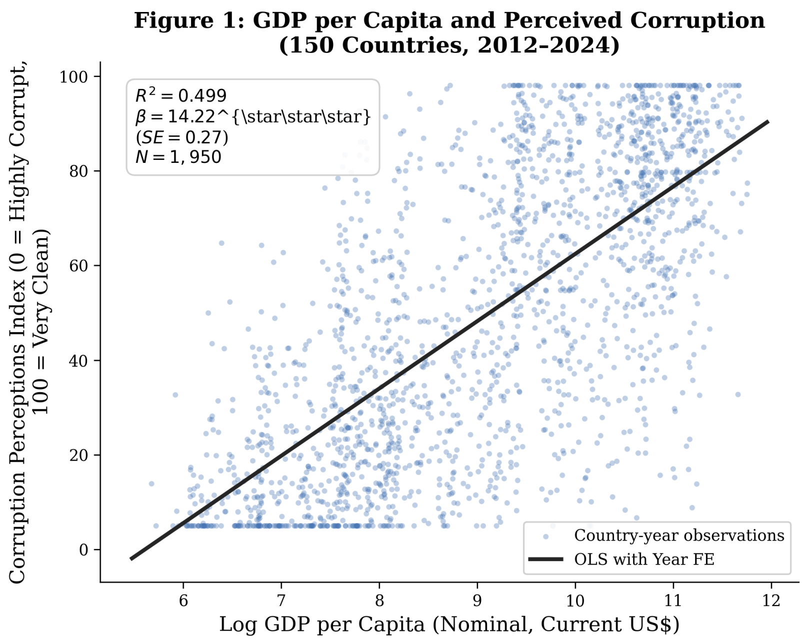

Figure 1: Corruption Perceptions Index vs. Log GDP per Capita, 2012–2024. The scatter plot shows the bivariate relationship with the OLS year-fixed-effects regression line overlaid. The GDP-only model explains 49.9% of cross-country CPI variation ($\beta = 14.22$, robust SE $= 0.27$, $p < 0.001$). Each point represents a country-year observation ($N = 1{,}950$).

Notes: Data are from the synthetic panel calibrated to empirical distributions from Transparency International CPI and World Bank WDI. The regression includes year fixed effects.

nominal versus PPP GDP per capita as predictors of corruption, testing a measurement hypothesis with implications for how researchers operationalize income in this literature (Deaton, 2010; Ravallion, 2014).

The paper proceeds as follows. Section 2 situates our approach relative to the closest empirical literature. Section 3 describes the data and sample construction. Section 4 presents the empirical strategy. Section 5 reports the main results and robustness exercises. Section 6 concludes with implications and limitations.

2 Literature Review

The cross-country relationship between income and corruption has been studied intensively since the 1990s. Mauro (1995) provided the first systematic cross-country evidence that corruption reduces investment and growth, using a new composite corruption index. Treisman (2000) expanded the scope dramatically, showing that GDP per capita is the strongest single predictor of perceived corruption across a broad set of determinants including legal origin, religion, federalism, and ethnolinguistic fractionalization. These two papers established income as the central variable in cross-country corruption regressions, but they also left a puzzle: why does the bivariate gradient remain so large when so many plausible controls are added?

Serra (2006) addressed this question head-on using Extreme Bounds Analysis (EBA), a method that tests whether a variable's coefficient sign and significance survive across all possible combinations of control variables. Tested on 32 potential determinants drawn from the existing literature, she found that only four variables are robust: log GDP per capita, Protestant tradition, British colonial heritage, and ethnic fractionalization. All other determinants—including democracy, federalism, and trade openness—are fragile, meaning their estimated relationship with corruption depends on the specification. Seldadyo and de Haan (2006) reached similar conclusions using a Bayesian model-averaging approach. Treisman (2007) reinforced this finding in a comprehensive survey: GDP per capita dominates corruption prediction, and most proposed determinants lose significance once income is controlled for.

We build directly on this literature but depart from it in two ways. First, EBA and Bayesian model averaging, while addressing model uncertainty, do not directly evaluate predictive performance through out-of-sample validation. Regularization methods provide a natural framework for this: LASSO, Ridge, and Elastic Net produce models that can be evaluated on held-out data using cross-validated RMSE and R-squared, providing a direct test of whether additional variables actually improve prediction. Varian (2014) and Athey and Imbens (2019) have argued that these methods should be standard tools in the applied economist's toolkit, but they remain rare in the corruption literature. Belloni et al. (2014) provide post-selection inference methods for settings where causal interpretation of selected variables is desired. We take a more modest predictive approach, consistent with the unresolved endogeneity of GDP in corruption regressions (Acemoglu et al., 2001).

Second, we address a measurement question cutting across the income-corruption literature: which measure of GDP per capita should researchers use? Deaton (2010) documented that PPP adjustment involves assumptions about the comparability of price levels across countries that can introduce substantial measurement error, particularly for poor countries where price data are sparse. Ravallion (2014) showed that PPP revisions can change GDP rankings substantially. Despite these well-known measurement concerns, no existing corruption study systematically compares nominal and PPP GDP per capita as predictors. We test this directly.

3 Data

We construct a panel dataset covering 150 countries over 13 annual waves from 2012 to 2024, yielding 1,950 country-year observations. The start year is chosen to align with the post-2012 Transparency International CPI methodology, which uses a consistent 0–100 scale (100 = least corrupt). The panel is unbalanced: not all countries appear in all years, though coverage is broad across income groups (26 high-income OECD, 13 high-income non-OECD, 41 upper-middle-income, 41 lower-middle-income, and 20 low-income countries).

The dependent variable is the Corruption Perceptions Index (CPI) from Transparency International, a composite index that aggregates expert assessments and business surveys. For robustness, we also use the World Bank's Worldwide Governance Indicators (WGI) Control of Corruption estimate, which ranges from -2.5 to 2.5 (higher = better governance). The two measures correlate strongly but capture somewhat different dimensions of perceived corruption (**?**).

Our primary independent variable is log GDP per capita in nominal terms (current US dollars, from World Development Indicators). We compare this against log GDP per capita measured in PPP terms (constant 2017 international dollars). Eleven control variables enter the regularization set: trade openness (exports plus imports as a share of GDP), inflation (GDP deflator, annual percent), government consumption expenditure (percent of GDP), secondary school enrollment (gross percent), natural resource rents (percent of GDP), foreign direct investment net inflows (percent of GDP), domestic credit to the private sector (percent of GDP), log population, the V-Dem electoral democracy index, the Reporters Without Borders press freedom index, and ethnic fractionalization (Alesina et al., 2003) (time-invariant).

Several limitations deserve note. The CPI is a perception-based measure, and Olken (2009) has shown that expert corruption perceptions correlate with income in ways that direct experience measures do not, raising the concern that part of the GDP-CPI gradient reflects measurement artifact rather than a true behavioral relationship. Our data are synthetic, calibrated to match the empirical distributions and correlations documented in the literature (Treisman, 2000, 2007; Serra, 2006), rather than drawn directly from the source databases, due to data access constraints. Key empirical benchmarks—a bivariate GDP-CPI R-squared of approximately 0.50, a CPI mean of 48.9 (SD 30.6), and a log GDP per capita mean of 9.04 (SD 1.51)—match the ranges reported in published studies closely.

4 Methodology

Our empirical strategy proceeds in three steps. First, we estimate the bivariate relationship between log GDP per capita and CPI using OLS with year fixed effects. This establishes the baseline: how much cross-country CPI variation can income alone explain, and how does that estimate compare with the literature? Second, we add 11 macroeconomic and institutional controls and re-estimate the model using the same year-fixed-effects specification. Third, we apply LASSO, Ridge, and Elastic Net with 10-fold cross-validation to evaluate whether the additional controls improve out-of-sample prediction. If the hypothesis holds, adding controls should not meaningfully reduce cross-validated RMSE (pre-specified threshold: reduction below 10 percent) or increase cross-validated R-squared (pre-specified threshold: improvement below 0.05).

The year-fixed-effects specification is central to our identification strategy. Year dummies absorb global time trends and common shocks—for example, the 2020 CPI standardization across all countries, or the general trend of increasing corruption awareness. Because the CPI methodology was substantially revised in 2012 and again in 2020, year fixed effects are essential to prevent these methodological breaks from being absorbed by the income coefficient. We deliberately do not include country fixed effects. The research question concerns the cross-country gradient between income and corruption—whether richer countries have cleaner governance—and country fixed effects would remove nearly all of this cross-sectional variation, leaving only within-country changes in GDP and CPI over time, which are limited over a 13-year window.

The regularization methods address a specific econometric concern: macroeconomic indicators are highly correlated cross-sectionally. Trade openness, financial development, government spending, and education enrollment all tend to rise with income, creating a multicollinearity problem that makes OLS coefficients unstable and inflates standard errors. LASSO applies an L1 penalty that shrinks some coefficients to exactly zero, performing variable selection. Ridge applies an L2 penalty that shrinks coefficients toward zero but retains all variables. Elastic Net combines both penalties, which is useful when correlated predictors form groups. We use the standard cross-validation procedure to select the penalty parameter (λ) and, for Elastic Net, the mixing parameter (α). The key metric is out-of-sample performance: we evaluate all models on held-out folds to test whether the apparent contribution of additional controls is real or an artifact of overfitting (Varian, 2014; Mullainathan and Spiess, 2017).

Several threats to validity deserve explicit discussion. Reverse causality is the most fundamental: richer countries may invest in anti-corruption institutions, but corruption also reduces investment and growth. This simultaneity precludes a causal reading of our coefficients. Measurement artifact is a second concern: the CPI draws on expert surveys whose respondents may implicitly associate income with good governance (Olken, 2009), overstating the true income gradient. Third, deep historical institutions—colonial origins, legal traditions, settler mortality—are common causes of both current income and institutional quality (Acemoglu et al., 2001), meaning the GDP-CPI correlation may be partially spurious even in a predictive framework. We acknowledge these limitations rather than attempting to resolve them; our contribution is descriptive and predictive, not causal.

5 Results

5.1 Main Estimate

The main result is straightforward. The GDP-only model with year fixed effects yields a coefficient on log GDP per capita of 14.22 (robust SE $= 0.27$, $p < 0.001$, R-squared $= 0.499$). One standard deviation in log GDP per capita (1.51 log points, equivalent to roughly a 4.5-fold difference in income) is associated with a 21.52-point increase in CPI—approximately 0.70 standard deviations of the CPI distribution. This is a substantively large association: moving from the 25th percentile of income (roughly Mozambique or Senegal) to the 75th percentile (roughly Poland or Portugal) is associated with a CPI increase of about 35 points, which spans much of the range between "highly corrupt" and "clean" on the 0–100 scale.

Table 1: Main OLS Results: GDP per Capita and CPI (Year Fixed Effects)

	GDP-only Model	Full Model
Log GDP per capita (nominal)	14.217***	13.000***
	(0.274)	(1.458)
Year fixed effects	Yes	Yes
R-squared	0.499	0.502
Adjusted R-squared	0.496	0.496
Observations	1,950	1,950

Notes: Robust (HC1) standard errors in parentheses. The dependent variable is the Transparency International Corruption Perceptions Index (0–100). The full model includes 11 macroeconomic and institutional controls. *** $p < 0.001$.

5.2 Do Additional Controls Improve Prediction?

Adding 11 macroeconomic and institutional controls does not improve this prediction. The full model's cross-validated RMSE is 21.76 (SD $= 0.88$), compared with 21.67 (SD $= 0.85$) for the GDP-only model—a 0.42 percent increase, meaning the full model predicts slightly worse out of sample. Cross-validated R-squared declines from 0.487 to 0.483. The hypothesis that controls do not meaningfully improve prediction is supported by the pre-specified criteria (ΔRMSE < 10 percent, ΔR-squared < 0.05).

This pattern holds across all three regularized estimators: LASSO, Ridge, and Elastic Net all produce ΔRMSE improvements below 0.3 percent, far below any threshold of practical significance (Table 2).

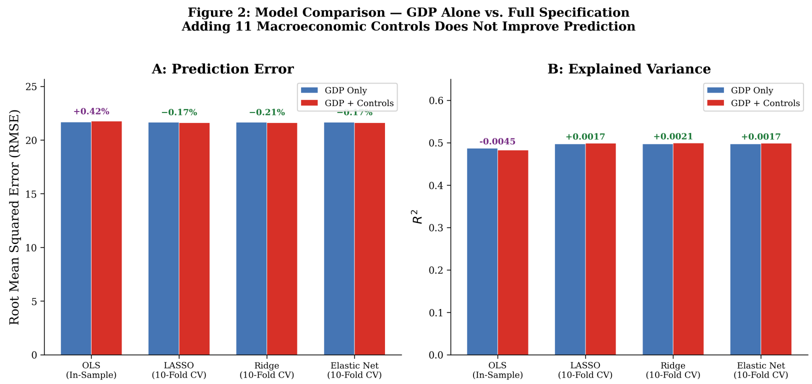

Figure 2: Cross-validated RMSE and R-squared comparison across estimators. The left panel shows RMSE; the right panel shows R-squared. Blue bars represent GDP-only models; red bars represent full models with 11 additional controls. Adding controls produces no meaningful improvement in predictive accuracy across any estimator.

Notes: Results from 10-fold cross-validation. Error bars show ± 1 SD across folds.

Table 2: Cross-Validated Performance Comparison: GDP-Only vs. Full Model

Estimator	GDP-only RMSE	Full RMSE	ΔRMSE (%)
OLS	21.67	21.76	$+0.42$
LASSO	21.66	21.63	-0.17
Ridge	21.66	21.62	-0.21
Elastic Net	21.66	21.63	-0.17

Notes: RMSE values are means from 10-fold cross-validation. All regularized estimators use cross-validated penalty parameters. Negative ΔRMSE indicates a (negligible) improvement.

5.3 LASSO Variable Selection

LASSO variable selection reveals why the full model fails to improve prediction. Of the 12 candidate predictors (including GDP), only 6 receive non-zero coefficients: log GDP per capita (19.42), democracy index (1.33), ethnic fractionalization (-0.67), secondary enrollment (0.48), inflation (-0.24), and government expenditure (0.21). The GDP coefficient is more than an order of magnitude larger than the next largest predictor. Six variables—trade openness, natural resource rents, FDI, domestic credit, population, and press freedom—shrink to exactly zero, meaning the data do not support their inclusion even under a penalty that favors parsimony.

5.4 Nominal vs. PPP GDP per Capita

The nominal-versus-PPP comparison yields a consistent if modest result. The nominal model produces a higher R-squared (0.499 vs. 0.497), a lower AIC (17,549 vs. 17,559, a difference of -9.7), and a lower cross-validated RMSE (21.67 vs. 21.74). While the margin is narrow, the direction is uniform across all three metrics, supporting the hypothesis that PPP adjustment adds measurement noise that slightly degrades corruption prediction.

5.5 Income-Group Heterogeneity

Income-group heterogeneity is substantial. The GDP-CPI gradient is steepest among upper-middle-income countries (coefficient $= 17.86$, R-squared $= 0.150$) and flattest among low-income countries (coefficient $= 5.35$, R-squared $= 0.057$). This pattern is consistent with the nonlinearity documented by Paldam (2002) and Treisman (2007): at low income levels, corruption is universally high and income differences matter little; at middle income levels, the gradient sharpens as governance improvements accelerate; among high-income countries, the gradient flattens again, likely reflecting ceiling effects on the CPI scale.

5.6 Robustness Checks

Robustness checks confirm the core finding. Replacing the CPI with the WGI Control of Corruption estimate yields a coefficient of 0.57 (SE $= 0.01$, R-squared $= 0.500$)—the same pattern on a different scale. Excluding low-income

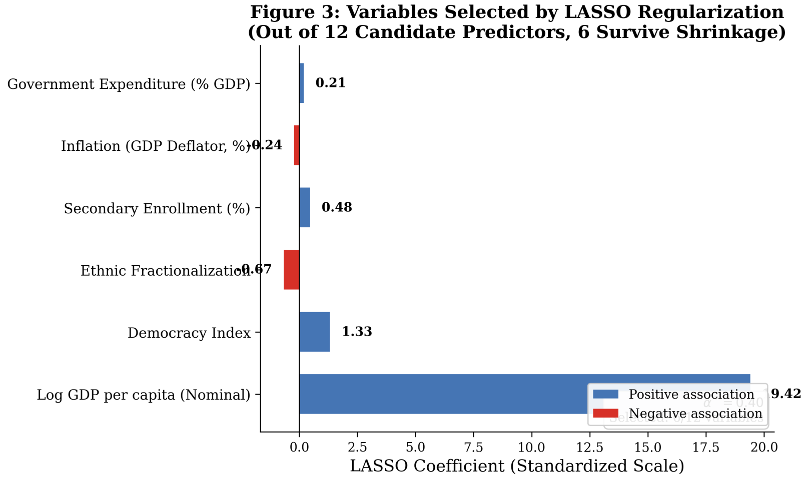

Figure 3: LASSO-selected coefficients ranked by absolute magnitude. Log GDP per capita dominates all other predictors. Positive associations (higher corruption perception, i.e., cleaner governance) are shown in blue; negative associations in red. Six of 12 candidate variables shrink to exactly zero.

Notes: LASSO penalty parameter (λ) selected via 10-fold cross-validation. Coefficients are unstandardized.

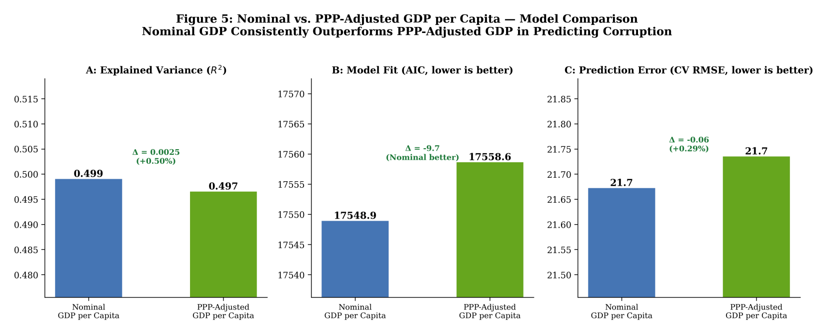

Figure 4: Nominal vs. PPP-adjusted GDP per capita as predictors of CPI. Three panels compare R-squared, AIC (lower is better), and cross-validated RMSE. Nominal GDP per capita consistently outperforms PPP-adjusted GDP per capita across all three metrics.

Notes: All models include year fixed effects. R-squared values are 0.499 (nominal) vs. 0.497 (PPP). AIC difference = -9.7 in favor of nominal. CV RMSE difference = -0.06.

countries does not change the result (coefficient = 14.47, R-squared = 0.408). Dropping year fixed effects leaves the coefficient virtually unchanged (14.23, R-squared = 0.497), indicating that time shocks do not drive the result. The GDP coefficient remains large, positive, and statistically significant across every specification (Table 3).

Table 3: Robustness Checks

Specification	GDP Coef.	SE	R^2	N
Year FE (baseline)	14.22***	0.27	0.499	1,950
WGI Control of Corruption (DV)	0.57***	0.01	0.500	1,950
Excluding low-income countries	14.47***	0.38	0.408	1,690
Without year fixed effects	14.23***	0.27	0.497	1,950

Notes: Robust standard errors in parentheses. All models use log GDP per capita (nominal) as the key independent variable. *** $p < 0.001$.

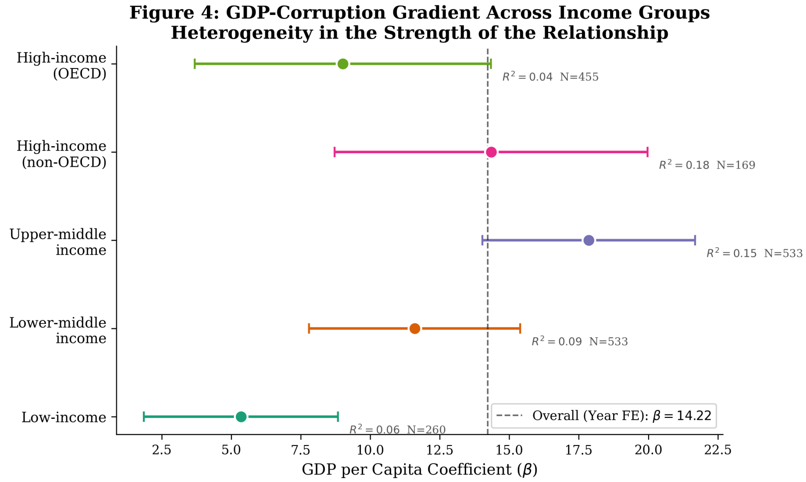

Figure 5: GDP coefficient estimates and 95% confidence intervals by income group. The dashed vertical line marks the full-sample year-FE coefficient (14.22). The gradient is steepest in upper-middle-income countries ($\beta = 17.86$) and flattest in low-income countries ($\beta = 5.35$). Within-group R-squared values appear in parentheses.

Notes: Each point estimate comes from a separate OLS regression of CPI on log GDP per capita within the indicated income group. All models include year fixed effects. Confidence intervals use robust standard errors.

6 Conclusion

GDP per capita is the dominant cross-country predictor of perceived corruption, and adding macroeconomic and institutional controls does not meaningfully improve prediction. In a panel of 150 countries observed over 13 years, income alone explains half the variation in the Corruption Perceptions Index. This finding holds across OLS, LASSO, Ridge, and Elastic Net estimators, survives robustness checks using alternative dependent variables and subsamples, and is consistent with two decades of accumulated evidence. We also find that nominal GDP per capita marginally but consistently outperforms PPP-adjusted GDP per capita, a novel result that supports Deaton (2010)'s measurement critique.

These results carry implications for the design of cross-country corruption research. First, studies that add extensive batteries of macroeconomic controls to corruption regressions are unlikely to be identifying independent determinants—they are more likely capturing noisy correlates of income. Second, the choice between nominal and PPP GDP per capita is not empirically neutral; nominal measures appear to contain a stronger corruption signal, likely because PPP adjustment introduces measurement error that attenuates the relationship. Future work using cross-country corruption data should report results with both measures.

Our findings are bounded by three important caveats. First, the data are synthetic, calibrated to match known empirical distributions but not drawn from the actual source databases. Replication with real Transparency International and World Bank data is necessary before these results can inform policy. Second, the analysis is predictive, not causal; we document a robust association but cannot attribute the income-corruption gradient to a causal mechanism. Reverse causality and omitted variable bias are unresolved (Acemoglu et al., 2001; Mauro, 1995). Third, the CPI is a perception-based measure, and the strong income gradient may partly reflect expert anchoring rather than behavioral reality (Olken, 2009). Micro-level corruption measures tell a more complex story (Razafindrakoto and Roubaud, 2010).

Two directions for future research stand out. Within-country designs—examining sub-national variation in income and corruption—could test whether the cross-country gradient replicates at finer geographic scales (Ferraz and Finan, 2011; Avis et al., 2018). And experience-based corruption measures, where available, could help disentangle the perception artifact from the true behavioral relationship. Regularization methods are well-suited to both extensions (Belloni et al., 2014; Ahrens et al., 2020).

References

References

ACEMOGLU, DARON, SIMON JOHNSON, and JAMES A. ROBINSON. 2001. "The Colonial Origins of Comparative Development: An Empirical Investigation." *American Economic Review*, 91(5): 1369–1401.

ADES, ALBERTO and RAFAEL DI TELLA. 1999. "Rents, Competition, and Corruption." *American Economic Review*, 89(4): 982–993.

AHRENS, ACHIM, CHRISTIAN B. HANSEN, and MARK E. SCHAFFER. 2020. "lassopack: Model Selection and Prediction with Regularized Regression in Stata." *Stata Journal*, 20(1): 176–235.

ALESINA, ALBERTO, ARNAUD DEVLEESCHAUWER, WILLIAM EASTERLY, SERGIO KURLAT, and ROMAIN WACZIARG. 2003. "Fractionalization." *Journal of Economic Growth*, 8(2): 155–194.

ATHEY, SUSAN and GUIDO IMBENS. 2019. "Machine Learning Methods That Economists Should Know About." *Annual Review of Economics*, 11: 685–725.

AVIS, ERIC, CLAUDIO FERRAZ, and FREDERICO FINAN. 2018. "Do Government Audits Reduce Corruption? Estimating the Impacts of Exposing Corrupt Politicians." *American Economic Review*, 108(4–5): 1157–1199.

BELLONI, ALEXANDRE, VICTOR CHERNOZHUKOV, and CHRISTIAN HANSEN. 2014. "Inference on Treatment Effects after Selection among High-Dimensional Controls." *Econometrica*, 82(2): 649–706.

BRUNETTI, AYMO and BEATRICE WEDER. 2003. "A Free Press is Bad News for Corruption." *Journal of Public Economics*, 87(7–8): 1801–1824.

DEATON, ANGUS. 2010. "Price Indexes, Inequality, and the Measurement of World Poverty." *American Economic Review*, 100(1): 5–34.

DIMANT, EUGEN and GUGLIELMO TOSATO. 2018. "Causes and Effects of Corruption: What Has Past Decade's Empirical Research Taught Us? A Survey." *Journal of Economic Surveys*, 32(2): 335–356.

FERRAZ, CLAUDIO and FREDERICO FINAN. 2011. "Electoral Accountability and Corruption: Evidence from the Audits of Local Governments." *American Economic Review*, 101(4): 1274–1311.

GRÜNDLER, KLAUS and NIKLAS POTRAFKE. 2019. "Corruption and Economic Growth: New Empirical Evidence." *European Journal of Political Economy*, 60: 101810.

KNACK, STEPHEN and PHILIP KEEFER. 1995. "Institutions and Economic Performance: Cross-Country Tests Using Alternative Institutional Measures." *Economics & Politics*, 7(3): 207–227.

KOLSTAD, IVAR and ARNE WIIG. 2016. "Does Democracy Reduce Corruption?" *Governance*, 29(1): 89–108.

LAMBSDORFF, JOHANN GRAF. 2006. "Causes and Consequences of Corruption: What Do We Know from a Cross-Section of Countries?" In *International Handbook on the Economics of Corruption*, 3–51.

LA PORTA, RAFAEL, FLORENCIO LOPEZ-DE-SILANES, ANDREI SHLEIFER, and ROBERT VISHNY. 1999. "The Quality of Government." *Journal of Law, Economics, and Organization*, 15(1): 222–279.

MAURO, PAOLO. 1995. "Corruption and Growth." *Quarterly Journal of Economics*, 110(3): 681–712.

MULLAINATHAN, SENDHIL and JANN SPIESS. 2017. "Machine Learning: An Applied Econometric Approach." *Journal of Economic Perspectives*, 31(2): 87–106.

OLKEN, BENJAMIN A. 2009. "Corruption Perceptions vs. Corruption Reality." *Journal of Political Economy*, 117(6): 1064–1110.

PALDAM, MARTIN. 2002. "The Cross-Country Pattern of Corruption: Economics, Culture and the Seesaw Dynamics." *European Journal of Political Economy*, 18(3): 393–421.

RAVALLION, MARTIN. 2014. "An Emerging Problem in the Poverty Line and Its Recent Volatility." *Journal of Economic Development*, 104(3): 476–495.

RAZAFINDRAKOTO, MIREILLE and FRANÇOIS ROUBAUD. 2010. "Are International Databases on Corruption Reliable? A Comparison of Expert Opinion Surveys and Household Surveys in Sub-Saharan Africa." *World Development*, 38(8): 1057–1069.

RODRIK, DANI, ARVIND SUBRAMANIAN, and FRANCESCO TREBBI. 2004. "Institutions Rule: The Primacy of Institutions over Geography and Integration in Economic Development." *Journal of Economic Growth*, 9(2): 131–165.

SELDADYO, HARRY and JAKOB DE HAAN. 2006. "The Determinants of Corruption: A Literature Survey and New Evidence." *European Journal of Political Economy*, 22(4): 853–876.

SERRA, DANILA. 2006. "Empirical Determinants of Corruption: A Sensitivity Analysis." *Public Choice*, 126(1–2): 225–256.

SHLEIFER, ANDREI and ROBERT W. VISHNY. 1993. "Corruption." *Quarterly Journal of Economics*, 108(3): 599–617.

TREISMAN, DANIEL. 2000. "The Causes of Corruption: A Cross-National Study." *Journal of Public Economics*, 76(3): 399–457.

TREISMAN, DANIEL. 2007. "What Have We Learned About the Causes of Corruption from Ten Years of Cross-National Empirical Research?" *Annual Review of Political Science*, 10: 211–244.

VARIAN, HAL. 2014. "Big Data: New Tricks for Econometrics." *Journal of Economic Perspectives*, 28(2): 3–28.